

Exercise 6 of module 1004D on "Theoretical spectroscopy for systems implying periodic boundary conditions"**Molecular properties****Wigner's 2n+1 rule**

Prove Wigner's 2n+1 rule for $n = 1$ by computing for the Hamilton operator $\hat{H}$,

$$\hat{H} = \hat{H}_0 + \lambda \hat{W}, \quad (1)$$

the perturbed energies to first and third order, E_1 and E_3 and show using the perturbed wave functions of first and second order that only the wave function to first order is needed to compute the energy to third order.

The Lagrange formalism

The Lagrangian $\mathcal{L}$ is defined as

$$\mathcal{L} = E + \bar{\lambda} e, \quad (2)$$

with the energy E , the constraints e and the multipliers $\bar{\lambda}$. Show that the second derivative of the energy with respect to the perturbation x can be written as

$$\epsilon^{(2)} = \mathcal{L}^{(20)} + 2\mathcal{L}^{(11)}\lambda^{(1)} + \mathcal{L}^{(02)}\left\{\lambda^{(1)}\right\}^2 \quad (3)$$

with

$$\mathbf{L}^{(mn)} = \left. \frac{\partial^{m+n} \mathcal{L}}{\partial^m x \partial^n \lambda} \right|_{x=0, \lambda=\lambda_*} \quad (4)$$

and

$$\lambda^{(1)} = \frac{d\lambda}{dx}. \quad (5)$$

Dalgarno's theorem

Assume that the second derivative of the energy depends on the coordinates x and y as well as on the amplitudes $\lambda(x, y)$. Based on the Lagrangian $\mathcal{L}$,

$$\mathcal{L} = E + \bar{\lambda} e, \quad (6)$$

compute the derivatives

$$\frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y}, \quad \frac{d}{dy} \frac{\partial \mathcal{L}}{\partial x}. \quad (7)$$

Compare the two derivatives.
Use the following nomenclature

$$E^{(klm)} = \frac{\partial^{k+l+m}}{\partial x^k \partial y^l \partial \lambda^m} E, \quad (8)$$

$$e^{(klm)} = \frac{\partial^{k+l+m}}{\partial x^k \partial y^l \partial \lambda^m} e, \quad (9)$$

$$\lambda^{(mn)} = \frac{\partial^{m+n}}{\partial x^m \partial y^n} \lambda, \quad (10)$$

$$\bar{\lambda}^{(mn)} = \frac{\partial^{m+n}}{\partial x^m \partial y^n} \bar{\lambda}. \quad (11)$$

Polarizability of the hydrogen atom in its ground state (Stark effect)

- Write down the electronic Schrödinger equation for the hydrogen atom in a constant electrical field $\mathbf{E} = E\mathbf{e}_z$. The wave function depends parameterically on E .
- Write down the equation for the first-order perturbation $\varphi^{(1)}(\mathbf{r})$. Which boundary conditions result from the normalization of $\varphi_E(\mathbf{r})$?
- Write down the equation for the static polarizability α .
- Solve the equation for $\varphi^{(1)}$ with the ansatz $\varphi^{(1)}(\mathbf{r}) = z(a + br)e^{-\beta r}$ (in atomic units) and compute α .

Magnetic susceptibility of noble gas atoms

Take a look at the Hamilton operator for a noble gas atom in a constant magnetic field $\mathbf{B}$. The magnetic field $\mathbf{B}$ is taken into account by substituting

$$\mathbf{p} \rightarrow \mathbf{p} + \frac{1}{c}\mathbf{A}(\mathbf{r}), \quad \mathbf{p} = \frac{1}{i}\nabla, \quad (12)$$

with $\mathbf{B}(\mathbf{r}) = \nabla \times \mathbf{A}(\mathbf{r})$.

- Show that the vector potential

$$\mathbf{A}(\mathbf{r}) = \frac{1}{2}(\mathbf{B} \times (\mathbf{r} - \mathbf{r}_0)) \quad (13)$$

is creating a constant magnetic field $\mathbf{B}$ for an arbitrary gauge origin $\mathbf{r}_0$.

- Show that the Hamilton operator $\hat{H}(\mathbf{B})$ is then given as

$$\hat{H}(\mathbf{B}) = \hat{H}_0 + \frac{1}{2c}\mathbf{B} \sum_k ((\mathbf{r} - \mathbf{r}_0) \times \hat{\mathbf{p}})_k + \frac{1}{8c^2} \sum_k (\mathbf{B} \times (\mathbf{r} - \mathbf{r}_0))_k^2, \quad (14)$$

with k denoting the number of electrons.

- Reformulate $\hat{H}(\mathbf{B})$ with $\mathbf{B} = B\mathbf{e}_z$ as

$$\hat{H}(\mathbf{B}) = \hat{H}_0 + B\hat{H}_1 + B^2\hat{H}_2. \quad (15)$$

- Verify the equation for the susceptibility of the ground state, given as

$$\chi = \langle \varphi_0 | \hat{H}_2 | \varphi_0 \rangle + \sum_{I>0} \frac{|\langle \varphi_0 | \hat{H}_1 | \varphi_I \rangle|^2}{E_0 - E_I}, \quad (16)$$

with (φ_j, E_j) representing eigen values and functions for $B = 0$.

- Show that the second term in χ vanishes if the gauge origin is at the position of the nucleus.